AMRITA VISHWA VIDYAPEETHAM
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Type 1: Research Project

Real-Time Cyber-Physical Threat Detection for Industrial Control Systems

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1. ICS/SCADA in Water Treatment

What are ICS and SCADA?

ICS (Industrial Control Systems) are systems used to monitor and control industrial processes, while SCADA (Supervisory Control and Data Acquisition) is a centralized system that supervises, collects data, and provides operator control over distributed field devices.

- Modern water treatment plants rely on ICS/SCADA to automatically control pumps, valves, and chemical dosing, ensuring safe and reliable drinking water.
- IT/OT convergence has brought previously isolated PLCs online: SCADA workstations, cloud dashboards, and remote vendor access now traverse shared enterprise networks.
- Adversaries are evolving from noisy ransomware to stealthy semantic/logic attacks that subtly alter setpoints and control logic (Stuxnet-style) while the HMI continues to display “normal” operation.

2. Need and Challenges: Detect Stealthy Logic Attacks Without Replacing OT

Needs:

- Water utilities require mechanisms to detect semantic/logic attacks on dosing, levels, and flow without rip-and-replace upgrades of existing PLC and SCADA installations.
- They need lightweight, external monitoring that understands the physics of the process for example, tanks cannot fill instantaneously and chemical concentrations cannot jump 100× in a single step.

Challenges:

- Hardware constraint: Legacy water-plant PLCs offer limited CPU and memory, making on-device cryptography and heavyweight AI agents impractical.
- Protocol weakness: Widely deployed Modbus/TCP provides no native authentication or integrity, so any host on the OT network can issue “valid” commands to a PLC.
- Latency constraint: Security controls must not disrupt sub-second control loops that protect pumps and tanks; detection must be near real-time and non-intrusive to process safety.

3. Review of Study of Chosen Problem

Condensed Literature Survey

Paper / Focus	Advantages	Limitations
Karacayılmaz et al. *AI-based IIoT* [1] (2024)	Adaptive to new threats; high accuracy using advanced AI models.	"Black-box" nature lacks explainability; high computational cost for legacy PLCs.
Murugesan et al. *Statistical Approach* [2] (2024)	Handles missing data effectively; mitigates early attack discrimination.	Limited to smart grids; not applicable to broader ICS environments.
Mohammed et al. *Field flooding* [3] (2023)	Specifically tailored for O&G infrastructure; effective against DoS.	Narrow scope (flooding only); may miss subtle Logic Injection attacks.
Barsha & Hubballi *State Transition*[4] (2024)	Detects semantic attacks by validating the *sequence* of operations.	Manual state modeling is complex; struggles with high-speed, concurrent states.
Athar et al. *Dual-Layer Attestation* [5] (2025)	Strong integrity verification; prevents firmware tampering.	High overhead for real-time loops; requires specific architecture changes.
Srivastava et al. *DDoS in SCADA* [6] (2024)	Critical protection against Availability attacks (DDoS) ensuring uptime.	Focuses on Network Layer (volumetric); misses Application Layer (logic) attacks.

- **Survey Scope:** We analyzed **6 key papers** focusing on AI-based detection, Statistical approaches, and Modbus protocol & SCADA vulnerabilities in Industrial Control Systems (ICS).

4. Gap Formulation

From Literature to Research Problem

Observed Limitation

Existing systems either introduce unacceptable latency / resource overhead on legacy PLCs, or they focus only on network-level anomalies and fail to detect when a compromised PLC is actively "lying" to the HMI with physics-plausible but malicious values.

Research Gap

Critical ICS in sectors like water treatment and Oil & Gas face stealthy cyber-physical attacks under strict real-time and safety constraints, but traditional IT security is too heavy for legacy PLCs and SCADA. There is a lack of a **lightweight, physics-aware intrusion detection system** that:

- Runs externally to legacy PLCs without firmware changes.
- Monitors controller logic and process states in real-time.
- Flags semantic / logic attacks that violate physical constraints.
- Preserves the tight latency requirements of safety-critical loops.

5. Problem Statement

Design a lightweight, physics-aware intrusion detection system for an Industry 4.0 water treatment plant that can reliably catch stealthy semantic/logic attacks on legacy PLCs without modifying OT hardware or violating real-time control constraints.

6. Objectives

1. To simulate and analyze **attacks in ICS networks** using the Modbus TCP/IP protocol, including register manipulation and unauthorized command injection.
2. To design a **lightweight monitoring architecture** capable of observing PLC logic behavior and process data without introducing significant overhead to real-time control operations.
3. To validate the system's ability to **detect physics-defying anomalies in real time** by identifying inconsistencies between PLC-reported values and expected physical process behavior.

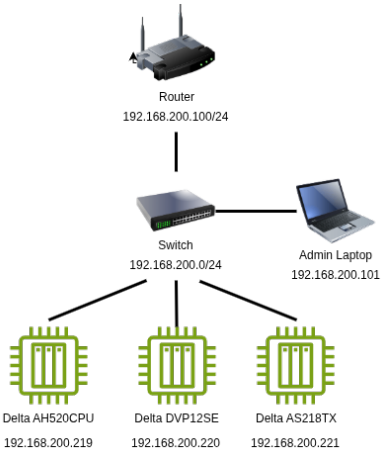
7. Deliverables

Expected Project Outputs

- **Open-source repository** containing ICS attack simulation scripts, Modbus communication utilities, and experimental test setups developed during Phase 1.
- **Mathematical state-transition model** describing the monitoring logic used to detect abnormal or illegal state transitions in industrial processes.
- **Final evaluation report** documenting experimental results, detection performance, and validation using standard ICS datasets.
- **Prototype monitoring framework** demonstrating real-time detection of attacks in an ICS environment.

8. ICS Testbed Topology

Hardware Architecture and Communication Flow



Lab Specifications

Control Nodes:

- **PLC 1:** Delta AH520CPU
- **PLC 2:** Delta DVP12SE
- **PLC 3:** Delta AS218TX

Network Backbone:

- 1x Edge Router (Delta DX-3021L9)
- 1x Network Switch (TP Link TL-SG3210)
- 1x Admin Laptop

Protocol Base:

- Modbus TCP/IP (Port 502)

9. Experimental Validation

From Dataset Simulation to Mitigation Testing

To evaluate our proposed anomaly detection system against real-world industrial conditions, we will follow a three-step execution plan:

Phase 1: SWaT Dataset Integration & Baselining

- **Action:** Simulate normal operational traffic and process physics from the **Secure Water Treatment (SWaT)** dataset directly onto our 3-PLC physical testbed.
- **Objective:** Establish a physical "ground-truth" baseline of expected sensor states, network flows, and actuator behaviors.

Phase 2: Threat Injection

- **Action:** Deploy our custom `pymodbus` exploitation framework against the active SWaT simulation.
- **Objective:** Recreate stealthy, application-layer attacks to generate a controlled, actively compromised OT environment.

Phase 3: Mitigation & Evaluation

- **Action:** Deploy the proposed real-time anomaly detection framework to monitor the compromised testbed.
- **Objective:** Measure the system's accuracy in identifying anomalies, track false-positive rates, and evaluate latency.

10. Feasible Attack Landscape

Simulated Threat Vectors mapped to MITRE ATT&CK for ICS

Based on the lab constraints (Modbus TCP/IP over a flat network), the following attack vectors are feasible. We focus on semantic, protocol, and logic manipulation rather than destructive firmware attacks.

Tactics Phase	Possible Attack Simulations in the Lab	MITRE ID
Discovery	**Network/Port Scanning:** Identifying live PLCs and open port 502/80.	T0846
	I/O Image Collection: Polling registers to map memory addresses.	T0877
Intrusion	**Session Hijacking:** Stealing Admin web cookies to bypass login.	T0859
Execution	**Digital Register Manipulation:** Overwriting Coils (e.g., forcing valves).	T0821
	Analog Register Manipulation: Overwriting Holding Regs (e.g., faking pressure).	T0836
Impact & Evasion	**Denial of Service (DoS):** Flooding the PLC to halt the control loop.	T0814
	Spoof Reporting Message: Intercepting/altering data sent to HMI.	T0856

11. Technical Evidence: Implemented Attacks

Live Testbed Exploitation

To validate the vulnerability of the ICS architecture, we successfully executed a multi-stage attack kill chain utilizing `pymodbus` and network exploitation tools.

Kill Chain Phase	Attack Action	MITRE ID	Execution & Result
1. Reconnaissance	Network & Port Scanning	T0846, T0843	Mapped OT topology and discovered open Modbus (502) and Web (80) ports without triggering network alarms.
2. Intrusion	Session / Cookie Hijacking	T0859	Intercepted active authentication cookies from the Admin laptop to bypass IT login mechanisms entirely.
3. Execution	Digital & Analog Manipulation (`pymodbus`)	T0821, T0836	Used Pymodbus to write to **Digital Coils** (forcing actuators) and **Analog Holding Registers** (altering pressure).
4. Impact	Denial of Service (DoS)	T0814	Flooded the PLC's network interface with high-frequency requests, exhausting CPU resources and halting the physical process.

12. Proposed Methodology: Hierarchical State-Based Monitoring

Dual-Layer Global Physical Validation

We propose a two-tier architecture that validates system integrity by correlating internal controller logic with global physical process laws.

Layer 1: Local State Model (PLC)

Focus: Logic Integrity & Sequencing.

- Monitors the **internal transitions** of individual PLCs by analyzing the transitions of the PLC state.
- **Detection:** Flags "Illegal Transitions" (e.g., jumping from *Idle* to *High Pressure* without the *Ramping* state).
- Catches code-level tampering and unauthorized logic changes.

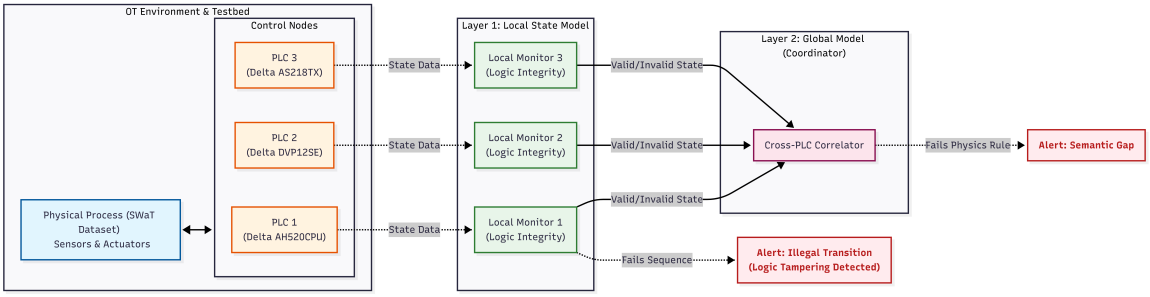
Layer 2: Global Model (Coordinator)

Focus: Process Physics & Cross-PLC Correlation.

- Monitors the **relationship** between multiple assets (e.g., Pump vs. Tank).
- Aggregates all the local state models to monitor the overall workflow.
- **Detection:** Flags "Semantic Gaps" where PLC reports "Normal" but the physics (flow/pressure) indicate a spike.

12. System Architecture

Hierarchical State-Based Monitoring Design



Conclusion: Research Direction & Impact

Real-Time Cyber-Physical Threat Detection for ICS

The Challenge

Industrial Control Systems face evolving threats that current defenses cannot handle:

- **Real-time constraints**: Detection must not disrupt safety-critical operations.
- **Legacy hardware**: Minimal CPU/RAM in existing PLCs and SCADA systems.
- **Stealthy attacks**: Multi-stage compromises that evade traditional network-based IDS.
- **Critical impact**: Breaches in water industry can cause catastrophic failures.

Expected Outcome

- Framework ready for adoption in Oil & Gas, water utilities, and power grids.
- Addresses critical research gap in real-time threat detection for legacy OT systems.
- Bridges advanced cybersecurity with practical ICS constraints.

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Thank You